

The Impact of Artificial Intelligence-Assisted Learning on Nursing Students' Ethical Decision-making and Clinical Reasoning in Pediatric Care

A Quasi-Experimental Study

Hyewon Shin, PhD, RN, Jennie C. De Gagne, PhD, DNP, RN, NPD-BC, CNE, ANEF, FAAN,
Sang Suk Kim, PhD, RN, Minjoo Hong, PhD, RN

The integration of artificial intelligence such as ChatGPT into educational frameworks marks a pivotal transformation in teaching. This quasi-experimental study, conducted in September 2023, aimed to evaluate the effects of artificial intelligence-assisted learning on nursing students' ethical decision-making and clinical reasoning. A total of 99 nursing students enrolled in a pediatric nursing course were randomly divided into two groups: an experimental group that utilized ChatGPT and a control group that used traditional textbooks. The Mann-Whitney *U* test was employed to assess differences between the groups in two primary outcomes: (a) ethical standards, focusing on the understanding and applying ethical principles, and (b) nursing processes, emphasizing critical thinking skills and integrating evidence-based knowledge. The control group outperformed the experimental group in ethical standards and demonstrated better clinical reasoning in nursing processes. Reflective essays revealed that the experimental group reported lower reliability but higher time efficiency. Despite artificial intelligence's ability to offer diverse perspectives, the findings highlight that educators must supplement artificial intelligence technology with strategies that enhance critical thinking, careful data selection, and source verification. This study suggests a hybrid educational approach combining artificial intelligence with traditional learning methods to bolster nursing students' decision-making processes and clinical reasoning skills.

KEY WORDS: Artificial intelligence, Critical thinking, Ethical standards, Nursing education, Nursing student

Author Affiliations: College of Nursing, Ewha Womans University, Seoul, Republic of Korea (Dr. Shin); School of Nursing, Duke University, Durham, NC, United States (Dr. De Gagne); and Red Cross College of Nursing, Chung-Ang University, Seoul, Republic of Korea (Dr. Kim); and College of Nursing, Institute of Health Sciences, Gyeongsang National University, Jinju, Republic of Korea (Dr. Hong).

The authors have disclosed that they have no significant relationships with, or financial interest in, any commercial companies pertaining to this article.

Corresponding author: Minjoo Hong, PhD, RN, College of Nursing, Institute of Health Sciences, Gyeongsang National University, Jinju 52727, Republic of Korea (mj1225@gnu.ac.kr).

This is an open-access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

Copyright © 2024 The Authors. Published by Wolters Kluwer Health, Inc.

DOI: 10.1097/CIN.0000000000001177

Integrating generative artificial intelligence (AI) technology such as ChatGPT into educational frameworks represents a notable evolution in teaching methodologies.¹ This evolution holds particular significance in nursing education, where the complexities of subjects such as child abuse necessitate innovative instructional strategies. Nurses need to use skilled relational practices to engage parents while ensuring children's safety and well-being, even in complicated situations. However, they often face challenges in balancing a focus on the child while maintaining a positive relationship with the parents.² These complexities require more innovative instructional strategies to adequately prepare nursing students for real-world scenarios. The evolving capabilities of generative AI to facilitate personalized, accessible, and comprehensive learning experiences are increasingly being acknowledged.³

Although generative AI has seen rapid acceptance and promising applications across various sectors, including health-care, where it is lauded for its ability to process clinical data and aid in decision-making,⁴⁻⁶ its integration into education has elicited debate. Concerns regarding academic integrity, potential biases, propagation of inaccurate information, and potential degradation of students' clinical reasoning and problem-solving abilities have been highlighted.^{3,7,8} However, benefits, including aiding faculty in the design of case studies and addressing challenging topics within nursing education, such as child abuse, cannot be overlooked. AI's ability to generate nuanced and complex case studies offers a valuable resource for educators, especially in scenarios where direct clinical experience is limited.¹

The interaction between millennials and AI technologies, especially chatbots, is significant. Arnold noted that 70% of millennials use chatbots, and more than half of those who have yet to use this technology have shown interest.⁹ This trend underlines the growing acceptance and integration of AI among younger generations, who are key players in the educational landscape. In the context of nursing, AI is increasingly utilized to collect data for behavior prediction models and risk assessment.¹⁰ A recent Korean review¹¹ highlights

that although many studies in South Korea focus on nursing students' awareness of AI, its application in education remains relatively limited. Most research centers on clinical nursing, with goals such as nurse turnover rates, staging pressure ulcers, and using robots for elderly care. Furthermore, the review underscores a significant need for research on the ethical implications of AI in nursing, which is an area of growing concern in Korean healthcare. Thus, discussions on AI in education have emphasized the need for a balanced approach that weighs its potential benefits against concerns related to privacy, security, and academic integrity.^{12,13} As AI's role in healthcare broadens, it is essential for health professionals and educators to understand how this technology can enhance patient care,^{12,13} particularly in specialized fields such as pediatric nursing that involve direct patient interactions.

Pediatric nursing care, with its ethical and professional complexities, stands to gain significantly from the strategic incorporation of AI. This potential is underscored by a recent study that explores how AI can improve the development of care plans—critical components of nursing practice—thereby enhancing both the efficiency and effectiveness of patient care.¹⁴ The study delves into AI's roles in evaluating patient data, enhancing communication, and identifying gaps in care plans, while also tackling significant ethical issues such as data privacy, fairness, and the balance between human and AI collaboration. It emphasizes the need for robust security measures, transparent algorithms, and clear accountability guidelines.¹⁴

Additionally, with the increasing adoption of AI technology in higher education and the growing research interest in their impact,^{15–17} there is a compelling need for empirical studies to assess these technologies' effects on nursing education. In response to this need, this research aims to evaluate the implications of integrating generative AI technology into nursing curricula, particularly in complex social contexts such as child abuse cases. It specifically assesses the differential impacts of AI-assisted learning versus traditional methods, aiming to provide educators with comprehensive insights into both the advantages and challenges of employing generative AI in nursing education.

METHODS

Design and Setting

This study employed a quasi-experimental design with experimental and control groups, using only a posttest to measure outcomes. The pediatric nursing course was divided into two classes, with a total enrollment of 110 students, of which 99 agreed to participate. To prevent coercion, the instructor (H.S.) of the course was unaware of who submitted the informed consent forms, ensuring that participation was voluntary, and confidentiality was maintained. Additionally, the group study report was not assessed as part of the academic performance for this class.

The decision to intervene was made after classes 1 and 2 students were written in an envelope and randomly selected by the instructor. Subsequently, class 1, with 52 students, was assigned to the experimental group, and class 2, with 47 students, was assigned to the control group. There were 11 subgroups within each group, consisting of four to six students each. Each student was assigned a code instead of a name to maintain anonymity. Group names and codes were noted upon the submission of the report.

The minimum sample size for each of the two groups was calculated as 26 based on a two-tailed test of the difference between two independent means with a 1:1 ratio, test power of 0.80, significance level of .05, and effect size of 0.80. Anticipating a 20.0% dropout rate, the study aimed for 66 participants (33 per group). However, in line with a previous research¹⁸ that targeted the entire class, we extended the opportunity to participate to every student attending the course.

Participants

The recruitment process was conducted by a research assistant who did not teach pediatric nursing. The research assistant introduced this study immediately after the second week of class and obtained informed consent from the students who agreed to participate voluntarily. The inclusion criteria were as follows: (1) nursing students, (2) enrollment in the Child Health Nursing I course, and (3) no prior case study experience in coursework utilizing AI. This study had no exclusion criteria. All 99 participants were female, as the university is a women-only university.

Interventions

Students attended a lecture on nursing care for child abuse in a pediatric nursing class. Throughout the class, students engaged in four group discussions on case studies, one of which was utilized for this research. The pediatric nursing class aimed to achieve several learning outcomes, two of which were selected for this study: the first focused on the professional and ethical standards of nursing care for children and their families, and the second on the application of nursing processes with critical thinking skills to address health problems in nursing situations targeting children and their families. Ethical standards involved considering ethical aspects in nursing care for children and their families, whereas nursing processes related to applying critical thinking skills to solve health issues targeting this demographic.

The instructor developed the initial drafts of the learning outcomes and evaluation rubrics, which were then reviewed and finalized by the research team. The first learning outcome criterion included five categories: (1) knowledge of professional standards, (2) understanding of ethical concepts, (3) communication of professional and ethical standards, (4) analysis of challenges and barriers, and (5) application of

ethical principles. The second applied nursing processes with critical thinking skills to solve health problems in nursing situations targeting children and their families, which included five categories: (1) application of nursing processes, (2) critical thinking skills, (3) integration of evidence-based knowledge and clinical reasoning, (4) collaboration and teamwork, and (5) reflection and improvement. In both scales, each item was evaluated on four levels: exemplary (3 points), proficient (2 points), developing (1 point), and not evident (0 points). The total score for a student was the sum of the scores for each section, with possible scores ranging from 0 to 15 in both categories.

Both the experimental and control groups were presented with a case of child abuse involving a 4-year-old child who was adopted by a couple and exhibited symptoms including a mild fever, dyspnea, and pain near the ribs; weighed 30% less than the average for their age; and had blurred scars on the back. To explore ethical and professional considerations, the experimental group used AI ChatGPT version 3.5 (OpenAI, San Francisco, CA, USA), and the control group used textbooks.

During the group activity, students spent approximately 30 minutes finding answers and discussing the results. They addressed several questions: in the nursing assessment phase, “What are the child’s problems?” and described two nursing assessments; in the nursing analysis phase, they described two priority nursing plans and interventions. Students also responded to four reflective questions about (1) what they learned from the group work using the case study, (2) what they needed to learn more after the case study, (3) the search methods or references used during the group study, and (4) their feelings or learnings from the search methods. After completing the group activity, the students engaged in a debriefing session to reflect on the ethical and professional considerations in nursing care for children and their families. This was a one-time group discussion.

Using student reports, two evaluators independently graded the scores based on two criteria (ethical standards and nursing processes) for the two groups, compared the results between the two criteria, and reached a consensus on any differences. The scores were similar in most cases, and an average score was calculated to represent the differences between the two evaluators.

Ethics

This study was approved by the institutional review board of University A (IRB no. 202309-0023-01). During the recruitment period, the purpose of the research, voluntary participation, confidential information, personal identity, and procedures were fully explained. Informed consent was obtained from each participant, who was informed that they could withdraw from the study at any time without any adverse consequences.

Statistic Data Analyses

SPSS (version 25.0; IBM Inc., Armonk, NY, USA) was used for the statistical analysis. The Mann-Whitney *U* test was used to assess differences between the two groups regarding ethical standards and nursing processes in child abuse cases. For the reflection paper, we counted each student’s answers and calculated the frequency and percentages.

RESULTS

Both participant groups consisted of the third-year nursing students who had not yet attended pediatric nursing classes or completed pediatric practicums. As such, they had comparable levels of experience and academic performance in pediatric nursing care. Table 1 shows significant differences between the experimental and control groups for “ethical standards” and “nursing processes,” with *P* values <.001 in each category. The control group demonstrated higher mean ranks (64.44 for “ethical standards” and 65.83 for “nursing processes”) than the experimental group (36.95 for “ethical standards” and 35.69 for “nursing processes”) (Table 1).

Table 2 reveals that in the first learning outcome (ethical standards), significant differences were observed in “understanding of ethical concepts,” “analysis of challenges and barriers,” and “application of ethical principles,” with the control group showing higher mean ranks than the experimental group (*P* < .001). In the second learning outcome (nursing processes), “critical thinking skills,” “integration of evidence-based knowledge and clinical reasoning,” and “reflection and improvement” showed significant differences (*P* < .001), with higher mean ranks in the control group (Table 2). According to the students’ reflection on the group activity, responses regarding what students learned from the child abuse case activities varied in both groups, with no single area receiving more than 50% of the responses. In the

Table 1. Comparison Between a Total Score of Ethical Standards and Nursing Processes Reports (N = 99)

Criteria	Group	n	Mean Rank	Sum of Ranks	Mann-Whitney U	P Value
Ethical standards	Exp.	52	36.95	1921.50	543.500	<.000
	Con.	47	64.44	3028.50		
Nursing processes	Exp.	52	35.69	1856.00	478.000	<.000
	Con.	47	65.83	3094.00		

Abbreviations: Con., control group; Exp., experimental group.

experimental group, the most learned aspect was recognizing the symptoms and signs of child abuse (n = 24, 46.15%), followed by the importance of considering complex situations (n = 21, 40.38%) and communication appropriate to the child's developmental process (n = 20, 38.46%). Meanwhile, in the control group, the most noted was the need to consider the child's complex physical and emotional situation (n = 21, 44.68%), followed by the importance of early and rapid response (n = 18, 38.30%), and nurses should identify priorities in child abuse cases (n = 10, 21.28%).

When examining students' responses in both the control and experimental groups regarding areas requiring additional learning after the child abuse case activity, communication methods appropriate for the child's developmental stage, child abuse nursing intervention (coping order, reporting method, and related laws), and the various clinical symptoms and characteristics of child abuse were ranked in the top three, although in different orders. In both groups, child abuse nursing interventions were mentioned in more than 50% of the responses.

There were notable differences in the variety of additional resources used by students in both groups. In the experimental

group, a more limited range of resources was utilized; about 21% of the students referred to textbooks, whereas a smaller percentage—9%—turned to other resources, including class lecture notes and search engines like the Google or Korean online platform such as Naver. In contrast, more than 90% of the students in the control group utilized search engines Google or Naver and hospital Web sites. Around 40% of students used class materials, whereas about 30% consulted research papers. The reflective essay revealed that within the experimental group, more than 90% of the students perceived low reliability in using AI technology, and nearly half acknowledged gaining a broader perspective and perceived efficiency in time when using AI. Meanwhile, in the control group, approximately 80% of the students experienced a longer duration when utilizing textbooks, and 64% reported a significant amount of information and the subsequent need to process it (Table 3).

DISCUSSION

This study explored the impact of AI as an educational resource in nursing students' engagement with child abuse case studies. The control group exhibited strong performance in

Table 2. Comparison Between Subcategory Scores of Ethical Standards and Nursing Processes Reports (N = 99)

Criteria	Group	n	Mean Rank	Sum of Ranks	Mann-Whitney U	P Value
Learning outcome 1 (ethical standards): Explain the professional and ethical standards used when approaching nursing care for children and their families.						
Knowledge of professional standards	Exp.	52	47.09	2448.5	1070.5	.26
	Con.	47	53.22	2501.5		
Understanding of ethical concepts	Exp.	52	41.19	2142	764	<.001
	Con.	47	59.74	2808		
Communication of professional and ethical standards	Exp.	52	50.58	2630	1192	.812
	Con.	47	49.36	2320		
Analysis of challenges and barriers	Exp.	52	40.85	2124	746	<.001
	Con.	47	60.13	2826		
Application of ethical principles	Exp.	52	37.42	1946	568	<.001
	Con.	47	63.91	3004		
Learning outcome 2 (nursing process): Apply nursing processes with critical thinking skills to solve health problems in the nursing situation targeting children and their families.						
Application of nursing processes	Exp.	52	48.06	2499	1121	.455
	Con.	47	52.15	2451		
Critical thinking skills	Exp.	52	33.42	1738	360	<.001
	Con.	47	68.34	3212		
Integration of evidence-based knowledge and clinical reasoning	Exp.	52	34.78	1808.5	430.5	<.001
	Con.	47	66.84	3141.5		
Collaboration and teamwork	Exp.	52	48.35	2514	1136	.531
	Con.	47	51.83	2436		
Reflection and improvement	Exp.	52	39.46	2052	674	<.001
	Con.	47	61.66	2898		

Abbreviations: Con., control group; Exp., experimental group.
Significant differences ($P < .001$) between the experimental and control groups are in bold font, indicating areas where the control group outperformed the experimental group.

ethical standards and clinical reasoning when applying the nursing process. Conversely, the experimental group, utilizing AI, quickly presented multiple problem-solving perspectives. This section outlines strategies to maximize AI's educational benefits and addresses the associated challenges in nursing classes.

The control group's superior performance in applying ethical standards may reflect limitations in AI's capacity for human-like contextual understanding and adaptation, as discussed by Korteling et al.¹⁹ Their reliance on textbooks facilitated deep engagement with the material, effectively linking theoretical knowledge to real-life nursing observations and seamlessly integrating ethical principles into their analyses. Addressing ethical issues is inherently challenging, but the control group enhanced their learning through active group discussions. These discussions promoted the sharing and integration of diverse viewpoints, a process believed to enhance critical thinking skills.²⁰ Furthermore, the control group's confidence in textbook answers likely stemmed from their familiarity with these materials. Given that AI is a relatively new method for Korean nursing students, it is crucial to prioritize education on AI usage for academic purposes and provide practical examples of how AI can be applied to ethical health issues.

The reflective essays provided key insights into what students learned from the case study and the sources of information they utilized. Students in the experimental group described their need for a systematic approach to problem-solving, which included identifying the symptoms of an abused child, assessing the situation, and implementing appropriate communication strategies. However, although these students followed structured interventions, the reflective essays indicated that the role of nurses was not clearly emphasized during the learning process. As a result, these students tended to accept answers passively, often without engaging in thorough group discussions that are vital for deepening understanding. In contrast, the control group was more engaged in discussions that allowed them to view scenarios from a nurse's perspective, focusing on situational assessment and priority setting. Yet, they sometimes overlooked essential assessments, such as evaluating a child's physical conditions. The findings suggest that integrating AI more effectively at the practical stages of nursing education, rather than limiting its use to theoretical discussions, could significantly improve learning outcomes by fostering a more active and comprehensive engagement with the material.

Regarding the sources of information used, the experimental group primarily relied on AI-generated responses without adequately cross-verifying them with textbook knowledge or current research findings. This reliance aligns with findings from Habib et al,²¹ who observed a decline in students' creative thinking abilities when using AI. The ability of AI to

quickly generate diverse and fluent responses may inhibit deep thinking and critical evaluation among students, potentially leading to an overdependence on technology. This concern is echoed by Bahrini et al²² and Dwivedi et al,²³ who noted that excessive reliance on AI might diminish critical thinking and reflection. Specifically, students lacking sufficient clinical experience might overly depend on AI, struggling to assess the reliability of its answers. According to Bisdas et al,²⁴ most students viewed AI more as a supportive partner than a competitor, a perception more prevalent among tech-savvy and medical students. In contrast, the control group engaged more thoroughly with traditional materials, necessitating extended time to critically evaluate information. This rigorous approach to data selection and analysis contributed to their higher perceived critical thinking skills in addressing complex child abuse cases.

The observed dependence on AI underscores the need for rigorous verification of AI-generated information and enhanced data selection skills. Active participation in this process ensures the validity of AI data and guides students in critical evaluation techniques. This practice is essential not only for knowledge acquisition but also for upholding ethical standards in the use of AI.²⁵ Moreover, a balanced approach to managing the privacy, security, and academic integrity risks associated with AI use, alongside its potential benefits, is crucial.^{12,13} Such measures are vital for maintaining critical thinking and ethical integrity in nursing education.²⁶ Consequently, students must not only engage critically with AI-generated data but also ensure their analyses are ethically sound and well-supported by accurate references, thereby cultivating a responsible and ethical use of AI in their learning.

Reflective essays from this study highlighted both the limitations and benefits of AI. Although most respondents (92.31%) in the experimental group cited the low reliability of AI-generated data, as previously noted by Cooper²⁷ and Jeyaraman et al,²⁸ and a study by Ngo et al²⁹ reported only a 32% success rate in generating correct answers, highlighting concerns about fidelity in academic settings.^{8,12} Despite these challenges, the benefits of AI, including systematic responses, new perspectives, and time efficiency, were evident. These positive findings echo the results of a prior study where participants' perceived usefulness scores for ChatGPT distinctly exceeded the median value.³⁰ Instructors are encouraged to inform students about these advantages and disadvantages before implementing AI and to guide them in verifying AI-sourced information with credible references, such as textbooks and research papers.

Additionally, the experimental group students demonstrated a broader approach to managing cases of child abuse, outlining five steps in nursing interventions, which include protecting the child, providing emergency care, documenting the event thoroughly, supporting the family through education, and

Table 3. Comparison of Reflection Paper (N = 99)

What Did They Learn From This Case Study? (1-2 Things)			
Experimental Group	n (%)	Control Group	n (%)
Signs and characteristics of child abuse	24 (46.15)	Considering complex conditions of child abuse (physical, emotional)	21 (44.68)
Considering complex conditions of child abuse (physical, emotional)	21 (40.38)	Importance of early response	18 (38.30)
Appropriate communication for a child's developmental stage	20 (38.46)	Nurses should identify priorities in child abuse cases	10 (21.28)
Assessing child-parent interactions	13 (25.00)	Importance of child abuse simulation	9 (19.15)
Education to prevent child abuse provided to children and their guardians	9 (17.31)	Signs and characteristics of child abuse	9 (19.15)
Importance of early responses	9 (17.31)	Assessing child-parent interactions	7 (14.89)
Importance of child abuse simulation	9 (17.31)	Need for learning how to respond to suspected child abuse situations	7 (14.89)
Nurses should identify priorities in child abuse cases	5 (9.62)	Education to prevent child abuse provided to children and their guardians	5 (10.64)
Importance of abilities to utilize ChatGPT	5 (9.62)	Appropriate communication for a child's developmental stage	4 (8.51)
After this group study, what do we need to learn more?			
Experimental Group	n (%)	Control Group	n (%)
Various conditions and characteristics of child abuse (symptoms and signs)	33 (63.46)	Appropriate communication methods for a child's developmental stage	26 (55.32)
Nursing intervention for child abuse (response sequence, reporting method, law)	29 (55.77)	Nursing intervention for child abuse (response sequence, reporting method, law)	24 (51.06)
Appropriate communication methods for a child's developmental stage	18 (34.62)	Various conditions and characteristics of child abuse (symptoms and signs)	22 (46.81)
Normal physical and language development stage by child's age	9 (17.31)	Normal physical and language development stage by child's age	18 (38.30)
Accumulating experience through child abuse simulation	9 (17.31)		
Search methods or references used during the group work (multiple choices)			
Experimental Group	n (%)	Control Group	n (%)
ChatGPT3.5 (free version)	52 (100.0)	Textbooks	47 (100.0)
Textbooks	11 (21.15)	Class materials (PowerPoint)	19 (40.03)
Class materials (ppt)	5 (9.62)	Search engines (hospital, blog)	43 (91.49)
Search engines (hospital, blog)	5 (9.62)	Research papers	14 (29.79)
Question 6: What they felt/learned from the search methods (1-2 things; multiple choices)			
Experimental Group	n	Control Group	n
Low reliability	48 (91.31)	Takes a long time	38 (80.85)
Systematic answers and new perspectives	25 (48.08)	Vast amount of data	30 (63.83)
Time-efficient	23 (44.23)	Data selection process is required	30 (63.83)
Answers vary depending on the questioning methods	22 (42.31)	Low reliability when searching online	28 (59.57)
Convenience of data collection	18 (34.62)	Deep learning through knowledge and cases is possible	13 (27.66)
Data selection process is required (use as supplementary, simple answers)	15 (28.85)	High reliability when using textbooks	9 (19.15)
Answers vary depending on the questioning language (English/Korean)	13 (25.00)		
Others (felt skeptical of the capabilities of AI)	5 (9.62)		

% = respondents/total participants in each group (experimental or control group).

engaging a multidisciplinary team involving legal and social services. This contrasts with the control group, which primarily focused on immediate nursing processes and contacting

the police, omitting the broader, multidisciplinary approach. This discrepancy suggests that AI can offer more detailed, context-specific information, which enhances

the depth of student responses to complex scenarios.⁴ AI's capacity to reduce search times³¹ and efficiently process large data sets to reveal patterns and relationships¹² further supports its utility as an independent learning resource that tailor responses to individual needs,⁴ underscoring its potential to significantly enhance science education.²⁸

Finally, evaluations of the lessons learned and additional learning needs identified by both AI and traditional learning groups revealed similar needs for further development in age-appropriate communication and nursing interventions. This similarity indicates that, regardless of the learning method, essential nursing skills such as age-appropriate communication are recognized as crucial by all students. The use of group discussions further helped align students' understanding and perceptions of the learning objectives, showcasing the value of collaborative learning environments.

Nursing Implications

The integration of AI learning technology into nursing education in Korea underscores the urgent need to incorporate such technologies in nursing curricula, despite challenges such as limited classroom time. Resources such as ChatGPT enable nursing students to access diverse perspectives and resources, which significantly enhance their critical thinking and data evaluation skills. To facilitate this, it is crucial for nursing education professionals and nursing associations in Korea to develop new AI-focused courses that offer both practical training and ethical guidance on using AI effectively.³²

The curriculum for these new courses should not only introduce nursing students to the fundamentals of AI but also cover its practical applications in clinical decision-making and the management of electronic health records.³³ Moreover, these courses need to address the ethical and legal implications of AI, ensuring that nursing students are well-prepared to navigate the complexities of AI in healthcare settings. Topics should also include an understanding of big data, AI concepts, algorithms, machine learning models, and deep learning, to provide a comprehensive background that supports the advanced use of AI in nursing practice.^{32,33}

Given the rapid expansion of big data and its implications for healthcare, AI systems and chatbots are becoming indispensable resources not only for academic researchers but also for clinical practitioners. The limited but growing body of research on AI's applications in education demonstrates a significant interest in exploring how AI can transform nursing education and practice. This increasing focus reflects the potential of AI to not only enhance educational outcomes but also improve clinical efficiencies and patient care.³⁴

Limitations

This study's design, employing cross-sectional data from participants in a single class at one university, limits the

generalizability of the findings to broader populations. Furthermore, to avoid potential discomfort, we did not collect individual data or conduct preactivity and postactivity surveys to assess students' awareness, knowledge, or attitudes toward ChatGPT, considering the group-based nature of the class activity. For future research, we recommend collecting basic demographic and educational background information on participants to enrich the analysis. Additionally, expanding the study to include various nursing specializations beyond pediatric care will provide a more comprehensive understanding of AI's impact across different areas of nursing education. This expanded scope will allow for a more nuanced analysis of AI technology such as ChatGPT in diverse contexts and learning environments.

CONCLUSION

Integrating traditional educational methods with AI technologies enables students to examine a broader range of patient cases from multiple perspectives, thereby enriching their educational experience. It is essential to guide the use of AI resource within educational frameworks, ensuring that ethical considerations are carefully addressed. AI technology facilitates efficient access to diverse viewpoints on nursing challenges. Nevertheless, reliance on AI should not diminish the fundamental need for critical thinking, careful data selection, and rigorous source verification by students. Merging AI with traditional learning approaches is crucial for enhancing nursing students' decision-making skills and ensuring the reliability of the information they utilize. This balanced approach aims to develop well-rounded nursing professionals equipped to navigate the complexities of healthcare with both technological proficiency and critical analytical skills.

References

1. De Gagne JC. Values clarification exercises to prepare nursing students for artificial intelligence integration. *International Journal of Environmental Research and Public Health*. 2023;20(14): 6409. doi:10.3390/ijerph20146409.
2. Lines MLE, Grant JM, Hutton A. 'How can we work together?' Nurses using relational skills to address child maltreatment in Australia: a qualitative study. *Journal of Pediatric Nursing*. 2020;54: 1–9.
3. Dilmegani C. Generative AI ethics: Top 6 concerns. Research AIMultiple. Web link. 2023. <https://research.aimultiple.com/generative-ai-ethics/>
4. Gilson A, Safranek CW, Huang T, et al. How does ChatGPT perform on the United States medical licensing examination? The implications of large language models for medical education and knowledge assessment. *JMIR Medical Education*. 2023;9(1): e45312. doi:10.2196/45312.
5. Goodman RS, Patrinely JR, Osterman T, Wheless L, Johnson DB. On the cusp: considering the impact of artificial intelligence language models in healthcare. *Med*. 2023;4(3): 139–140. doi:10.1016/j.medj.2023.02.008.
6. Kalla D, Smith N. Study and analysis of ChatGPT and its impact on different fields of study. *International Journal of Innovative Science and Research Technology*. 2023;8(3).
7. Archibald MM, Clark AM. ChatGPT: what is it and how can nursing and health science education use it? *Journal of Advanced Nursing*. 2023;79(10): 3648–3651. doi:10.1111/jan.15643.

8. Sallam M. ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. *Healthcare (Basel)*. 2023;11: 887. doi:10.3390/healthcare11060887.
9. Arnold A. How chatbots feed into millennials' need for instant gratification. *Forbes*. 2018. <https://www.forbes.com/sites/andrewarnold/2018/01/27/how-chatbots-feed-into-millennials-need-for-instant-gratification/?sh=44e22e5a3675>.
10. Ha JY, Park HJ. Keyword network analysis and topic modeling of news articles related to artificial intelligence and nursing. *Journal of Korean Academy of Nursing*. 2023;53(1): 55–68. doi:10.4040/jkan.22117.
11. Hong M, Shin H, Pi J. Artificial intelligence on nursing: a scoping review. artificial intelligence on nursing: a scoping review. *The Journal of the Convergence on Culture Technology (JCCT)*. 2024;10(2): 311–322. doi:10.17703/JCCT.2024.10.2.311.
12. Kooli C. Chatbots in education and research: a critical examination of ethical implications and solutions. *Sustainability*. 2023;15(7): 5614. doi:10.3390/su15075614.
13. O'Connor S. Open artificial intelligence platforms in nursing education: tools for academic progress or abuse? *Nurse Education in Practice*. 2022;66: 103537. doi:10.1016/j.nepr.2022.103537.
14. Nashwan AJ, Abujaber AA, Abujaber A. Nursing in the artificial intelligence (AI) era: optimizing staffing for tomorrow. *Cureus*. 2023;15(10): e47275. doi:10.7759/cureus.47275.
15. Ayers JW, Poliak A, Dredze M, et al. Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. *JAMA Internal Medicine*. 2023;183(6): 589–596. doi:10.1001/jamainternmed.2023.1838.
16. Han J, Park J, Lee H. Analysis of the effect of an artificial intelligence chatbot educational program on non-face-to-face classes: a quasi-experimental study. *BMC Medical Education*. 2022;22(1): 830. doi:10.1186/s12909-022-03898-3.
17. Taskiran N. Effect of artificial intelligence course in nursing on students' medical artificial intelligence readiness: a comparative quasi-experimental study. *Nurse Educator*. 2023;48(5): E147–E152. doi:10.1097/nne.0000000000001446.
18. Nam H, Jeong G. The impact of problem-based simulation education program on pediatric nursing student's academic achievement, academic efficacy, and clinical performance skills. *Journal of Korea Society for Simulation in Nursing*. 2017;5(2): 33–43.
19. Korteling J, van de Boer-Visschedijk GC, Blankendaal RA, Boonekamp RC, Eikelboom AR. Human-versus artificial intelligence. *Frontiers in Artificial Intelligence*. 2021;4: 622364. doi:10.3389/frai.2021.622364.
20. Mahdi OR, Nassar IA, Almuslamani HAI. The role of using case studies method in improving students' critical thinking skills in higher education. *International Journal of Higher Education*. 2020;9(2): 297–308. doi:10.5430/ijhe.v9n2p297.
21. Habib S, Vogel T, Anli X, Thorne E. How does generative artificial intelligence impact student creativity? *Journal of Creativity*. 2024;34(1): 100072. doi:10.1016/j.jyoc.2023.100072.
22. Bahrini A, Khamoshifar M, Abbasimehr H, et al. ChatGPT: applications, opportunities, and threats. *IEEE*. 2023;274–279. doi:10.1109/SIEDS58326.2023.10137850.
23. Dwivedi YK, Kshetri N, Hughes L, et al. "So what if ChatGPT wrote it?" multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*. 2023;71: 102642. doi:10.1016/j.ijinfomgt.2023.102642.
24. Bisdas S, Topriceanu C-C, Zakrzewska Z, et al. Artificial intelligence in medicine: a multinational multi-center survey on the medical and dental students' perception. *Frontiers in Public Health*. 2021;9: 795284. doi:10.3389/fpubh.2021.795284.
25. Welding Lee. "Half of College Students Say Using AI on Schoolwork Is Cheating or Plagiarism." *BestColleges*, 2023; <https://www.bestcolleges.com/research/college-students-ai-tools-survey/>.
26. Farhi F, Jeljeli R, Aburezeq I, Dweikat FF, Al-shami SA, Slamene R. Analyzing the students' views, concerns, and perceived ethics about chat GPT usage. *Computers and Education: Artificial Intelligence*. 2023;100180. doi:10.1016/j.caeai.2023.100180.
27. Cooper G. Examining science education in chatgpt: an exploratory study of generative artificial intelligence. *Journal of Science Education and Technology*. 2023;32(3): 444–452. doi:10.1007/s10956-023-10039-y.
28. Jeyaraman M, Ramasubramanian S, Balaji S, Jeyaraman N, Nallakumarasamy A, Sharma S. ChatGPT in action: harnessing artificial intelligence potential and addressing ethical challenges in medicine, education, and scientific research. *World Journal of Methodology*. 2023; 13(4): 170–178. doi:10.5662/wjm.v13.i4.170.
29. Ngo A, Gupta S, Perrine O, Reddy R, Ershadi S, Remick D. ChatGPT 3.5 fails to write appropriate multiple choice practice exam questions. *Academic Pathology*. 2023;11(1): 100099. doi:10.1016/j.acpath.2023.100099.
30. Kang SR, Kim SJ, Kang KA. Awareness of using chatbots and factors influencing usage intention among nursing students in South Korea: a descriptive study. *Child Health Nursing Research*. 2023;29(4): 290–299. doi:10.4094/chnr.2023.29.4.290.
31. de Souza LL, Fonseca FP, Martins MD, et al. ChatGPT and medicine: a potential threat to science or a step towards the future? *Journal of Medical Artificial Intelligence*. 2023;6. doi:10.21037/jmai-23-70.
32. Jeong GH. Artificial intelligence, machine learning, and deep learning in women's health nursing. *Korean Journal of Women Health Nursing*. 2020; 26(1): 5–9. doi:10.4069/kjwhn.2020.03.11.
33. Paranjape K, Schinkel M, Panday RN, Car J, Nanayakkara P. Introducing artificial intelligence training in medical education. *JMIR Medical Education*. 2019;5(2): e16048. doi:10.2196/16048.
34. Chen X, Xie H, Zou D, Hwang G-J. Application and theory gaps during the rise of artificial intelligence in education. *Computers and Education: Artificial Intelligence*. 2020;1: 100002. doi:10.1016/j.caeai.2020.100002.